PCB Design Workshop

Assignment 2

Smart Desk Assistant (Environment Monitor + Pomodoro Timer)

Task

Design a schematic and PCB layout in EasyEDA for a smart desk device that monitors room conditions and helps with focused study sessions using the Pomodoro method. This is an open design. Choose your own components and plan your own layout.

Note: *The components mentioned in brackets below are just examples to guide you. You are free to use these, or any other suitable component of your choice, as long as the function is achieved.*

What the Device Should Do

- **Sense the environment**: measure temperature, humidity, and light level *(e.g. DHT11 + LDR, or BME280)*
- **Show a display**: use a display that shows both the Pomodoro timer and the environmental sensor values *(e.g. 0.96" OLED, 1.3" OLED, or color TFT)*
- **Have 3 buttons**: START, STOP/PAUSE, RESET
- **Give alerts**: buzzer sound when a study session ends; red LED turns on if temperature is too high; green LED turns on when plugged into external power
- **Microcontroller**: use a microcontroller that supports Wi-Fi or Bluetooth, so the device can send timer/sensor data wirelessly. It's preferred to use a bare-metal microcontroller chip directly on your PCB, rather than a ready-made module/dev-board like Arduino Nano or ESP32 DevKit. *(e.g. ESP32-WROOM chip, ESP8266 chip)*
- **Show battery level**: measure and display remaining battery % *(e.g. simple voltage divider, or a fuel gauge IC)*
- **Run on two power sources**: plugged-in power or battery, with an ON/OFF switch and a voltage regulator *(e.g. USB-C/barrel jack + JST battery connector + AMS1117 regulator)*

Board Requirements

Board size	70-90 mm × 50-70 mm (max)
Minimum trace width	30 mil
Layers	Single-layer or double-layer; your choice
Components	SMD, THT, or a mix of both

You are free to choose the exact components, pin assignments, and layout — this assignment tests your ability to design a complete working system on your own, not just replicate a fixed reference. No code is needed — only the circuit design (schematic + PCB) will be evaluated.

Submit 3 Screenshots

1. Schematic diagram from EasyEDA
2. PCB layout (2D, routed)
3. 3D view of the board

Deadline: 12 September, 8:00 AM (সকাল ৮ টা)